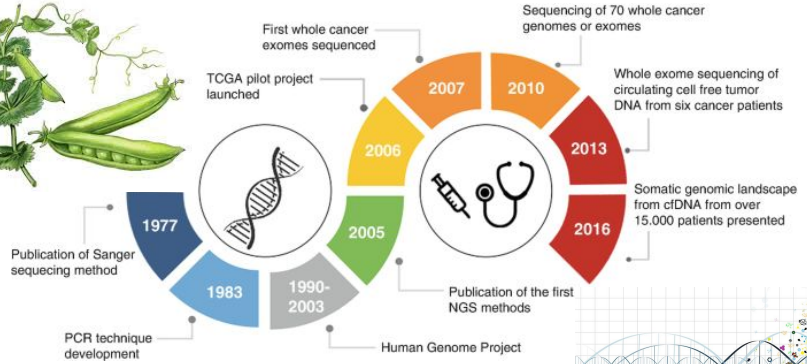
BME 105: Genetics in the Genomics Era

Lecture 1: Introduction and Mendel's Laws



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Assistant Professor

Course website on canvas.ucsc.edu

Some slides modified from Angela Brooks and Rohinton Kamakaka

Background + Lecture 1 Learning Objectives (BME 105)

Molecular biology topics, be able to:

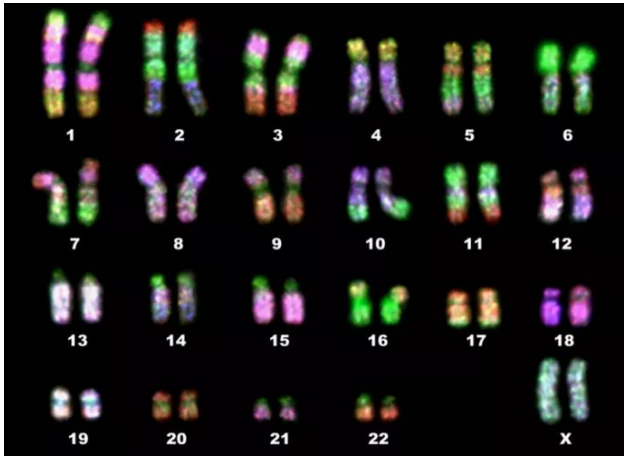
- Describe the central dogma, i.e., how information is transferred from genotype to phenotype.
- Understand the hierarchical levels of genetic organization: genomes → chromosomes → genes/non-coding DNA.
- Define and distinguish the terms: gene, allele, mutation.
- What is the "genetic code" and how is it read?
- Understand the definitions of DNA strands: coding, non-template, template. Which terms are equivalent?

** Please take the “Genetics Course Introductory Knowledge Survey” linked in the syllabus quiz on Canvas **

New genetics concepts, be able to:

- Articulate why model systems are important in genetics - what aspects are essential?
- Explain the difference between homozygous and heterozygous genotypes.
- Explain trait dominance and recessivity.
- Draw a Punnett square for a given pair of parental genotypes to determine the genotypes of their offspring
- Enumerate the combinatorial gametes produced in a dihybrid cross and use them in a Punnett square to determine the next generation's genotypes.
- Describe the Laws of Independent Segregation and Assortment (of Mendelian traits)

Fluorescently-labeled karyotype



What is a gene?

There are 23 chromosomes in humans

Each chromosome consists of one double stranded DNA molecule

Each chromosome has many genes (~100 (Y) to 1000s (autosomes))

A gene often (but not always) produces a protein.

Protein (and non-coding RNA) activity gives rise to a <u>phenotype</u>

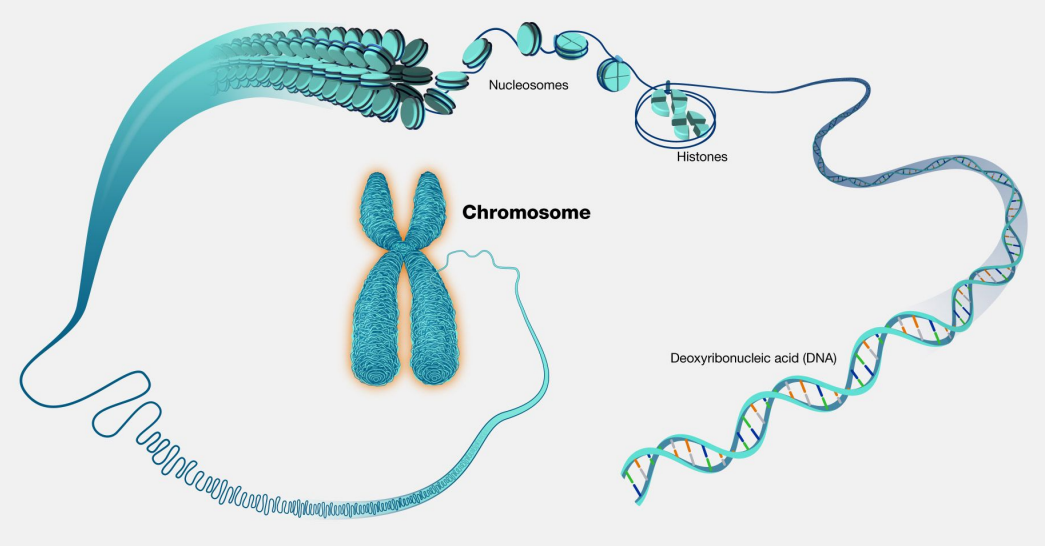
There are approximately 25,000-30,000 genes in almost **every cell of the human body**

There are two copies of each gene in each of these cells (Humans=diploids)

Each gene has <u>many</u> forms ~ alleles (there are only 2 alleles in any individual (diploid) human)

- Different alleles are caused by changes in the DNA sequence of the gene.
- These changes in DNA sequence are termed mutations.
- Some mutations change the encoded amino acid sequence.
- New protein sequences may have altered functions.
- Altered protein function can lead to the appearance of a new phenotype.

Chromosomal structure and gene organization



Darryl Leja, NHGRI

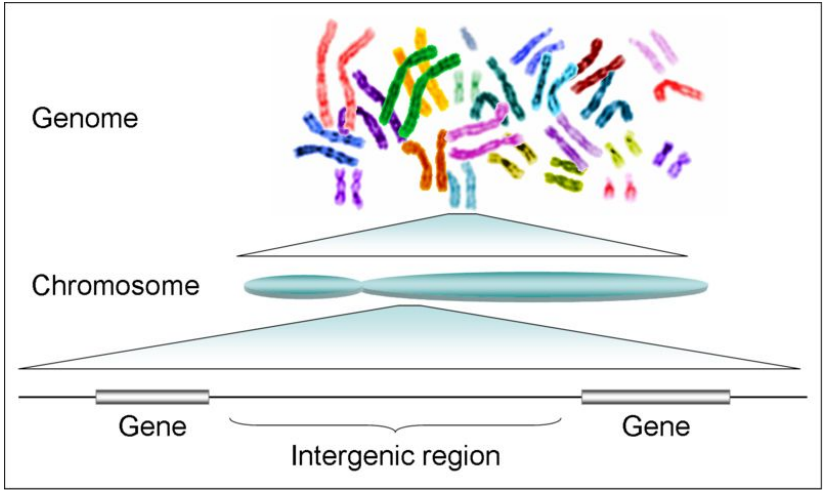
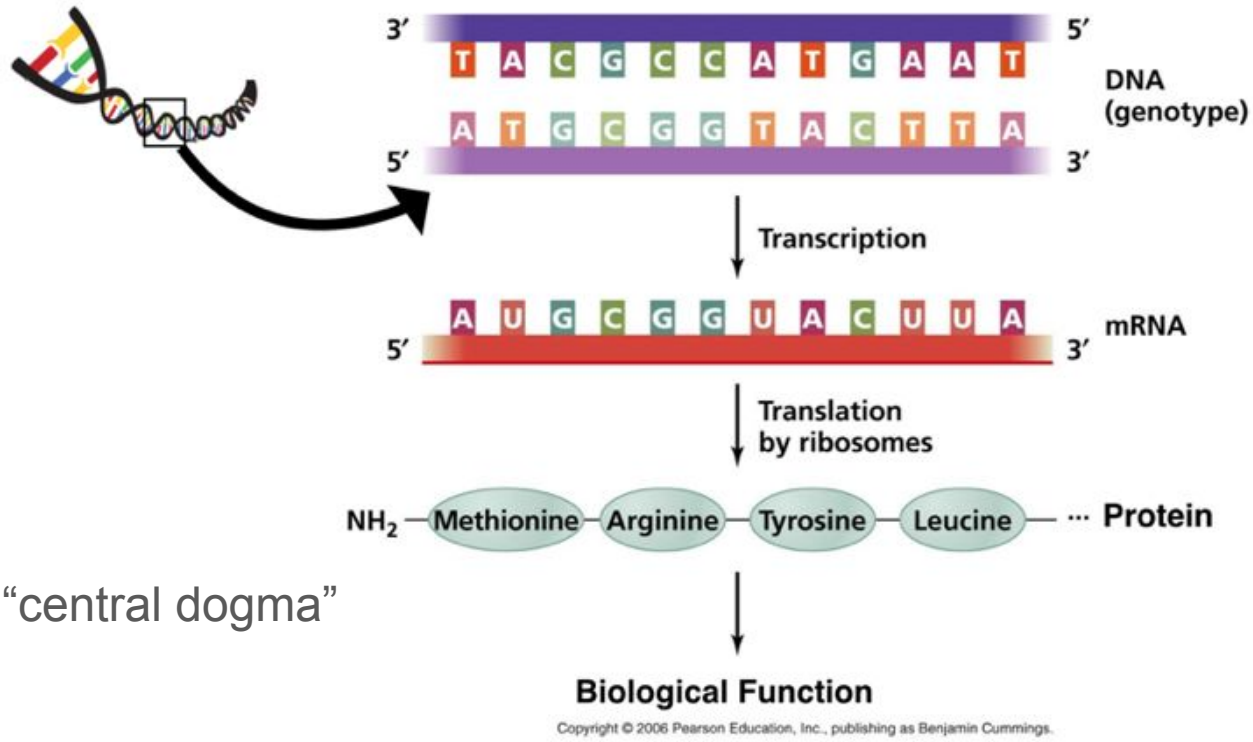


Image Credit: *Wikimedia Commons; User – Plociam*

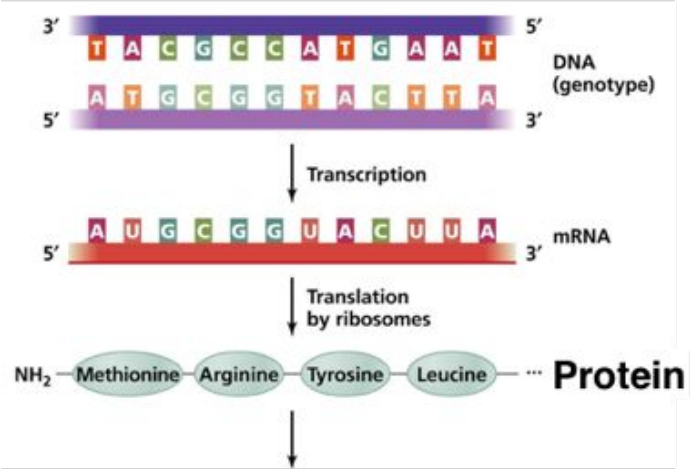
Eukaryotic information transfer: genotype to phenotype

(+ archaeal and bacterial)

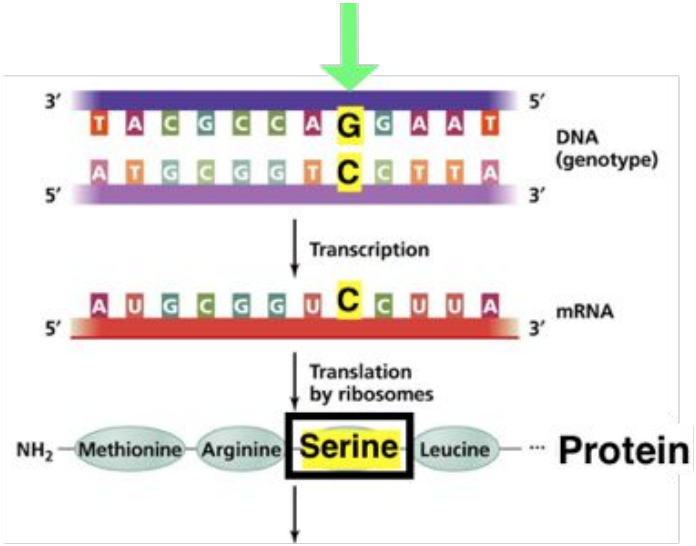


A.k.a., the “central dogma”

Alternative DNA sequences of a gene are termed alleles, which <u>may</u> encode an amino acid substitution

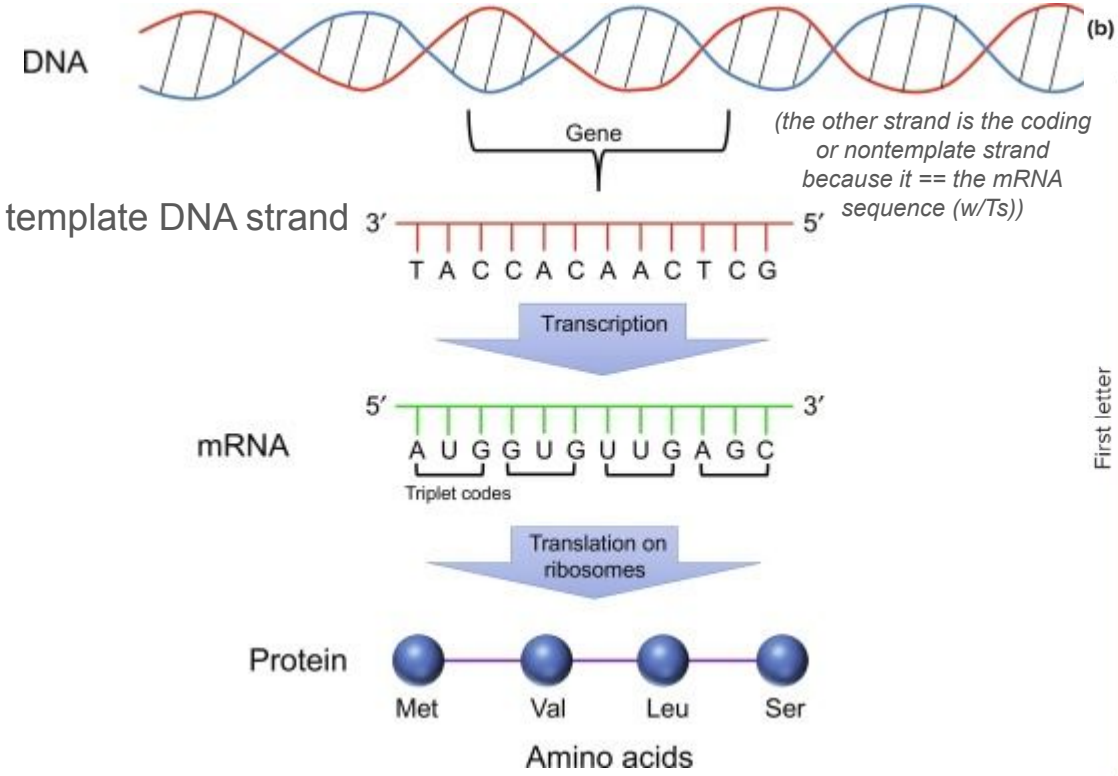


Biological function



Potentially modified or different biological function

Reading the genetic code



First letter	Second letter: U	Second letter: C	Second letter: A	Second letter: G	Third letter
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA Stop UAG Stop	UGU Cys UGC Cys UGA Stop UGG Trp	U C A G
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg	U C A G
A	AUU Ile AUC Ile AUA Ile AUG Met	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	U C A G
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	U C A G

Reading the genetic code: other genetic codes (are similar)

DNA

template DNA str

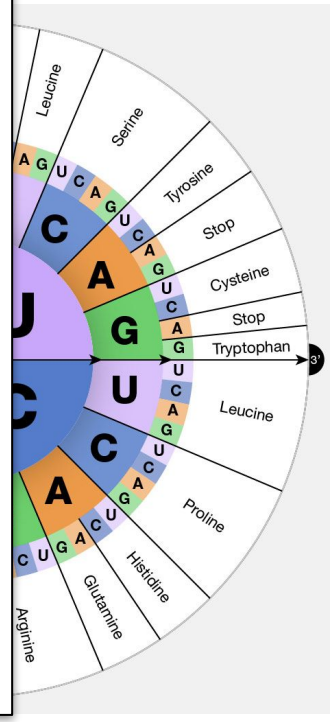
mRNA

Pr

Currently, genetic codes can be set independently for nucleus, mitochondria, plastids and hydrogenosomes. The current settings for each of these on the taxonomic tree can be viewed by the four buttons directly underneath the following code list.

The following genetic codes are described here:

- 1. The Standard Code ← Humans use these
- 2. The Vertebrate Mitochondrial Code ←
- 3. The Yeast Mitochondrial Code
- 4. The Mold, Protozoan, and Coelenterate Mitochondrial Code and the Mycoplasma/Spiroplasma Code
- 5. The Invertebrate Mitochondrial Code
- 6. The Ciliate, Dasycladacean and Hexamita Nuclear Code
- 9. The Echinoderm and Flatworm Mitochondrial Code
- 10. The Euplotid Nuclear Code
- 11. The Bacterial, Archaeal and Plant Plastid Code
- 12. The Alternative Yeast Nuclear Code
- 13. The Ascidian Mitochondrial Code
- 14. The Alternative Flatworm Mitochondrial Code
- 15. Blepharisma Nuclear Code
- 16. Chlorophycean Mitochondrial Code
- 21. Trematode Mitochondrial Code
- 22. Scenedesmus obliquus Mitochondrial Code
- 23. Thraustochytrium Mitochondrial Code
- 24. Rhabdopleuridae Mitochondrial Code
- 25. Candidate Division SR1 and Gracilibacteria Code
- 26. Pachysolen tannophilus Nuclear Code
- 27. Karyorelict Nuclear Code
- 28. Condylostoma Nuclear Code
- 29. Mesodinium Nuclear Code
- 30. Peritrich Nuclear Code
- 31. Blastocrithidia Nuclear Code
- 33. Cephalodiscidae Mitochondrial UAA-Tyr Code



Where to start?

What is needed to understand how phenotypic traits are inherited?

1. A breedable organismal system that
    a. Controls parental genotypes
    b. Produces large numbers of progeny (for statistics)

2. Observable, heritable phenotypic variation that is
    a. Measurable
    b. Follows a simple inheritance mode
        i. think: eye color (simple) vs height (complex)

Genetic Inheritance: Mendel and The Pea, *Pisum sativum*

Mendel chose the common garden pea to study patterns of inheritance.

This was a excellent choice as a model system because it:

1. Has many varieties (pure/true breeding types - *i.e.*, controlled genotypes)
2. Hermaphrodites that can self-pollinate and cross-pollinate
   - ➢ Individuals express both male and female organs
   - ➢ Self-fertilization is controllable by removing the anthers
3. Inexpensive to grow
4. Has a short generation time
5. Produces large numbers of progeny
6. Expresses <u>discrete</u> polymorphic phenotypes (alternative trait forms)
   - ➢ Opposed to continuous traits, fixed discrete phenotypes, etc.

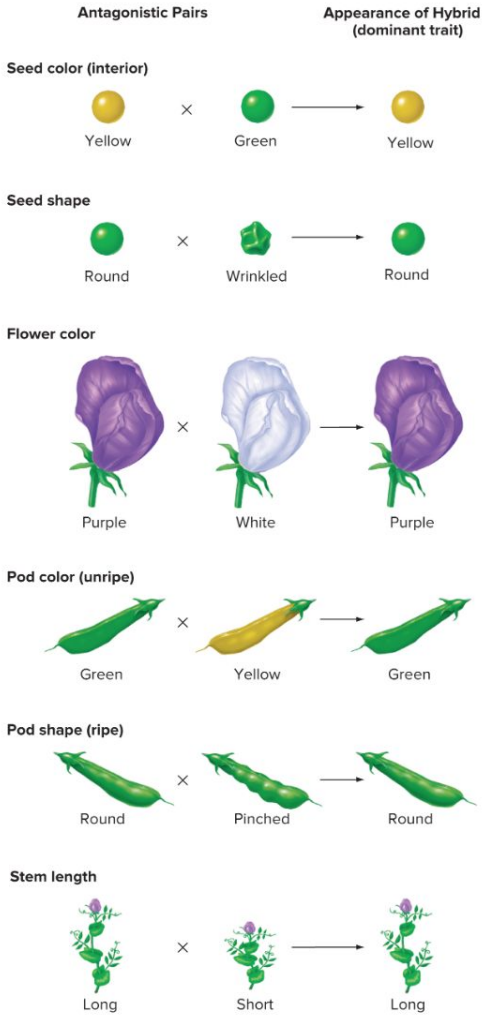
Mendel’s pea crosses

Started with pure-breeding (true-breeding) lines with antagonistic (differing) traits

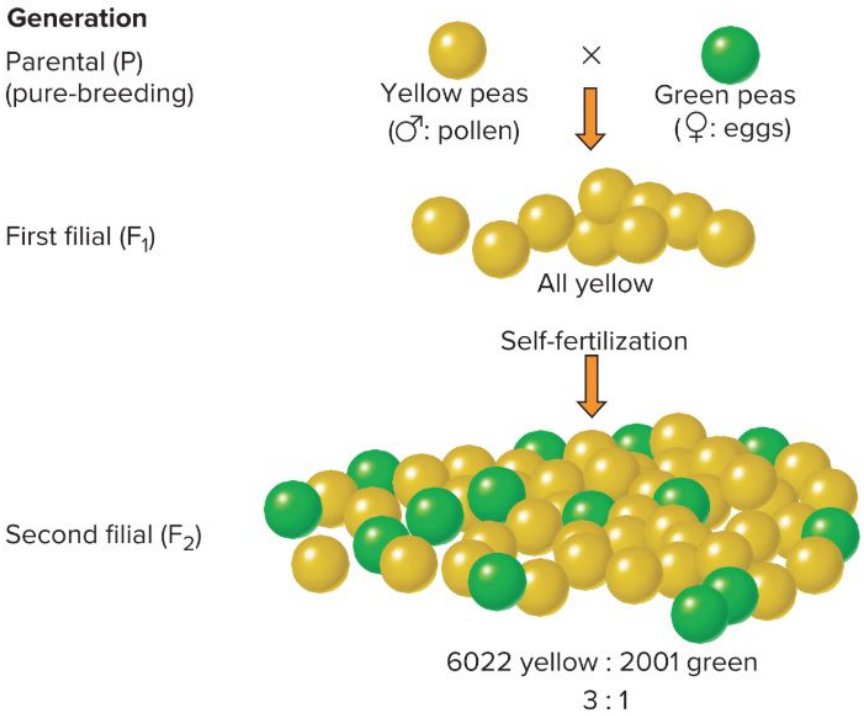
Crossed to make hybrids and observed a dominant trait

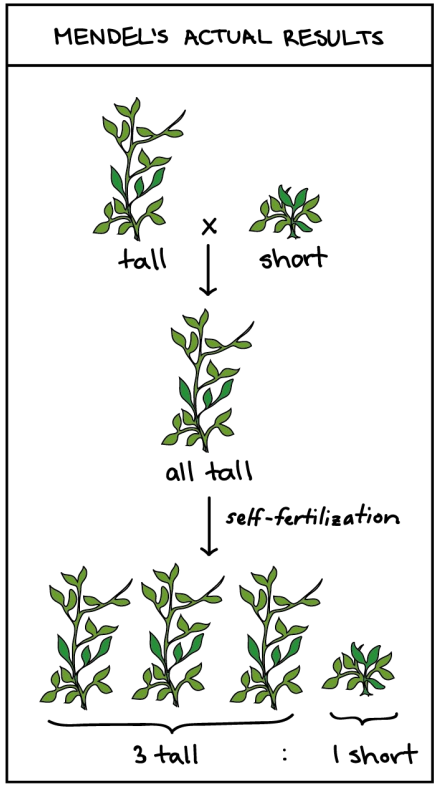
Performed reciprocal crosses to control for parental effects

Carefully controlled mating and maintenance conditions to ensure genetics underlied results



Mendel's investigations of the patterns of inheritance with <u>monohybrid crosses</u>:





- Both parents contribute equally
  → one allele from each
  (except, see organellar inheritance)

- Traits remain intact
  → recessive alleles reappear in F2s

Mendelian genetics terms

Dominant and recessive: All F1 seeds were yellow but when selfed the F2 produced some green seeds.

Mendel termed the trait that is expressed in the F1 as **dominant**

The trait that is hidden but re-expressed in the F2 as **recessive**

⇒ The F1 plants must contain factors for green and yellow since both reappear in the F2

With these assumptions the simplest model is that the F1 contains two hereditary factors (alleles)

One for green and another for yellow

Annotations:

Uppercase Y to represent the dominant yellow factor

Lower case y to represent the recessive green

⇒These conventions are used for other genes and alleles, with unique names.

Genotypes:

Homozygous - two of the same allele of a gene (e.g., YY or yy), encoded by a “homozygote”

Written Y/Y or y/y (modern convention)

Heterozygous - different alleles of a gene (e.g., Yy), encoded by a “heterozygote”

Written Y/y (modern convention)

Keys to Mendel's success

- He ran heritability experiments with all seven traits
- The ratios obtained were between 2.82:1 and 3.15:1 (vs 3:1 expected)
- Crosses involving only one single locus trait, e.g., seed color, is called a monohybrid cross.
- Protocol for success:
  - Generated a pure breeding line before starting the cross
  - Analyzed individual character traits
  - Counted progeny

Mendel's conclusions

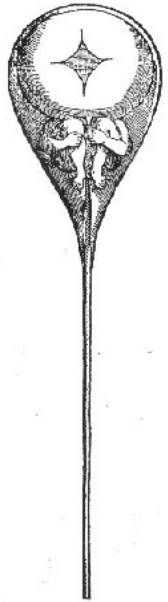
Although others were doing crosses at the time, Mendel's work was unique

1. In F1 all progeny showed only one of the two possible traits for all crosses

2. Character missing in F1, reappeared in F2
   ⇒ Traits did not blend in the offspring but were transmitted in a discrete fashion and remained unchanged.

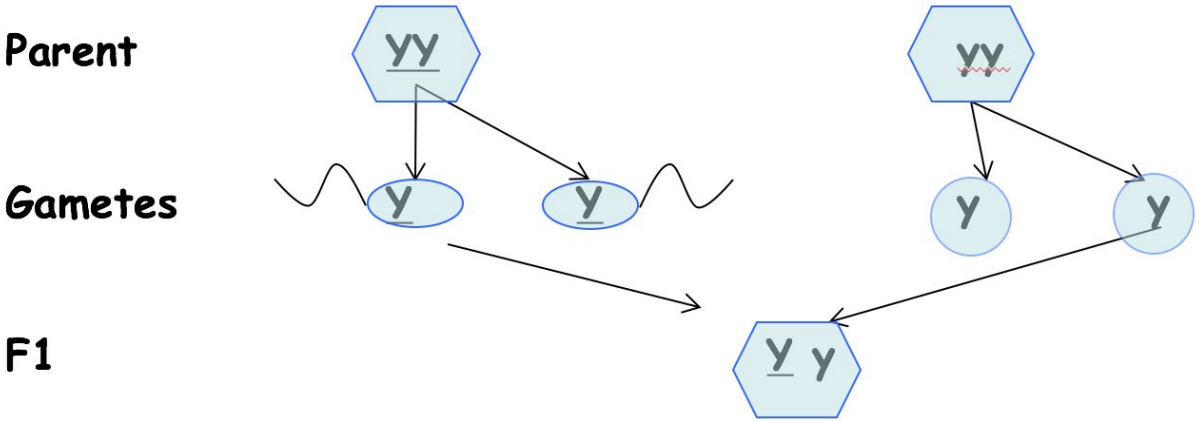
3. Males and females not important (i.e., autosomal traits)
   ⇒ Reciprocal crosses produced the same results, this indicated that <u>each parent makes an equal contribution to genetic makeup of the offspring</u>.
   ⇒ The sperm does not contain a homunculus.

Homunculus: A tiny fully-formed organism. Mendel's conclusions found this theory to be incorrect.



Mendel revealed gametic inheritance and the law of segregation

Mendel therefore reasoned that during gamete formation the paired factors separate and each gamete receives only one of the two factors.

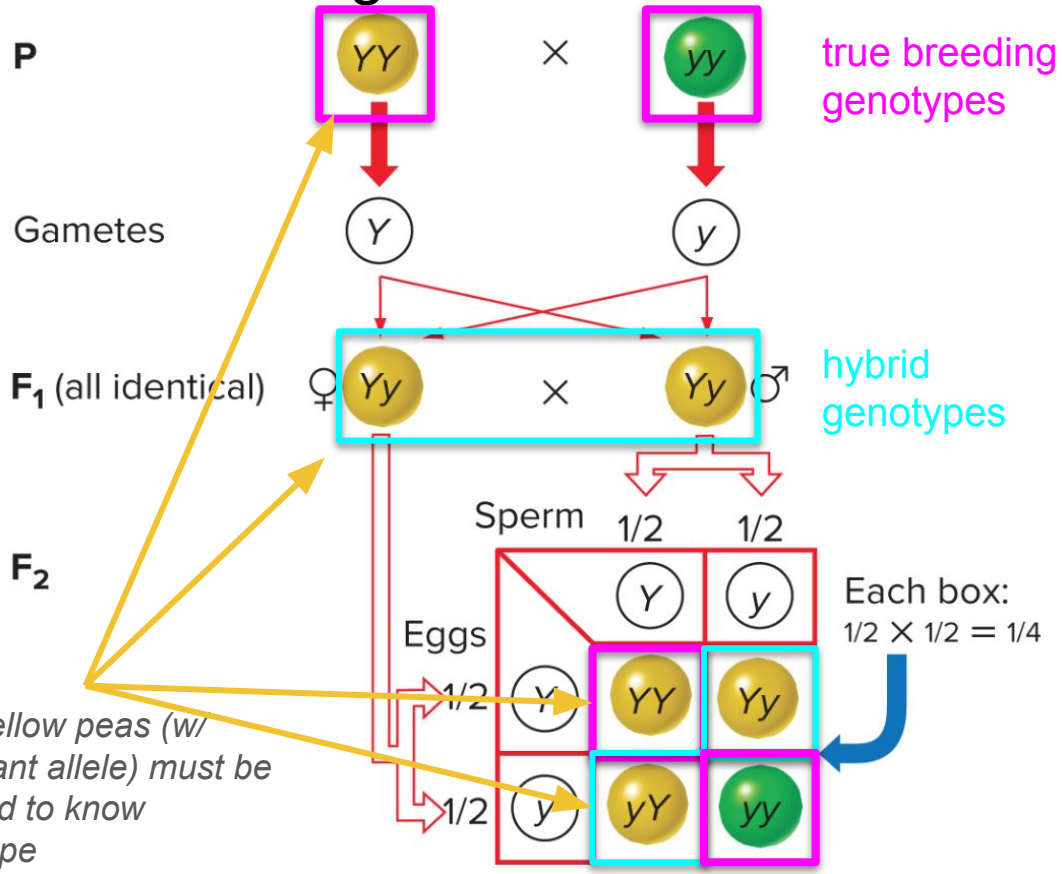


Sperm and egg then randomly combine to produce F2 progeny reconstituting two copies of each factor in the progeny

Punnett square cross visualization of gamete fertilization

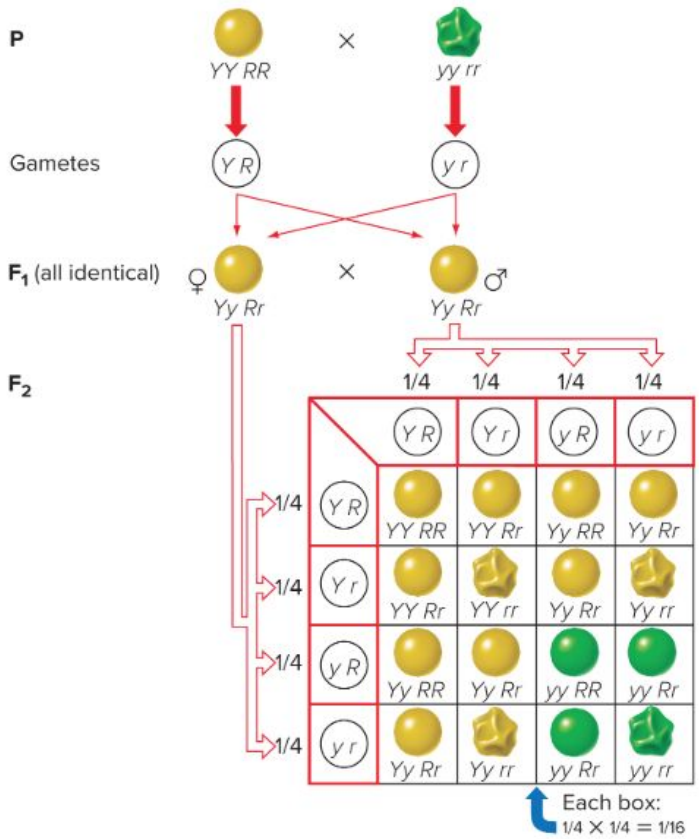
Enables:

- Designate dominant alleles
- Enumerate offspring genotypes
- Calculate probability of each genotype



***All yellow peas (w/ dominant allele) must be crossed to know genotype*

Dihybrid crosses reveal the Law of Independent Assortment



Law of independent assortment: During gamete formation, different pairs of alleles segregate independently of each other

Dihybrids are heterozygous for two genes

"Recombinant types" are new phenotypic combinations

→ not to be confused with "recombinant chromosomes", which we'll discuss soon (lecture 8)